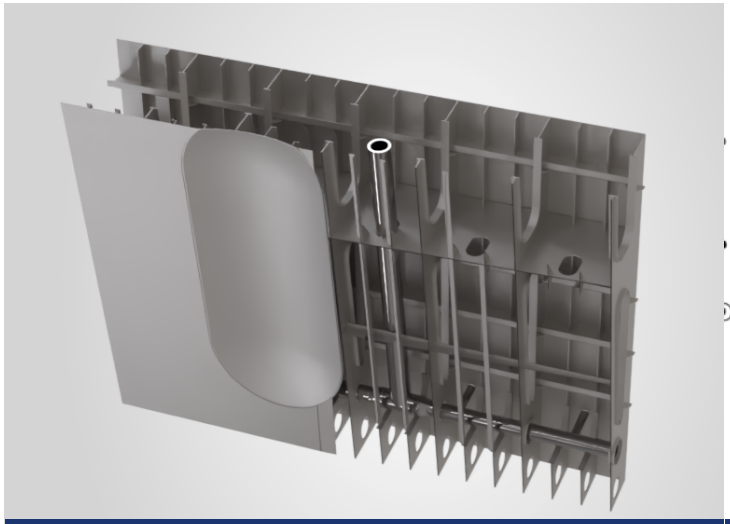
# Relevant Projects

Outlined below are a variety of projects that I have worked on. Projects and experiences include work done during my co-ops at Ontario Shipyards, Dometic and Takachar, design teams projects from UBC Thunderbikes and Concrete Toboggan, course related projects and personal projects.

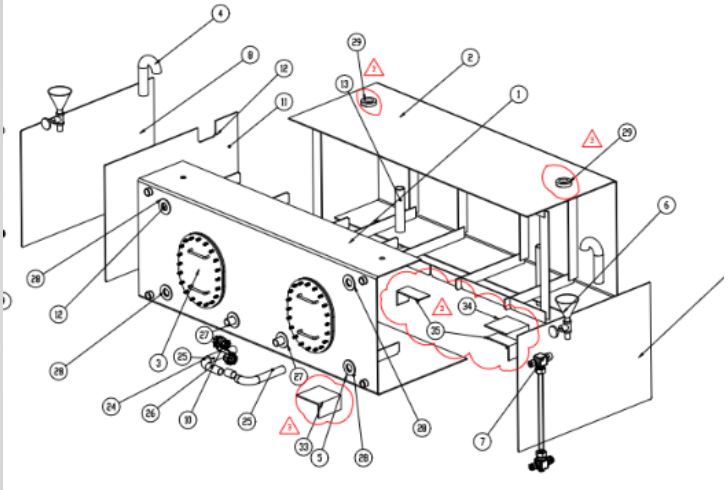
## Ontario Shipyards - Project Engineering Intern

*AutoCAD LT, SolidWorks, CNC Plasma Cutting, MIG Welding*

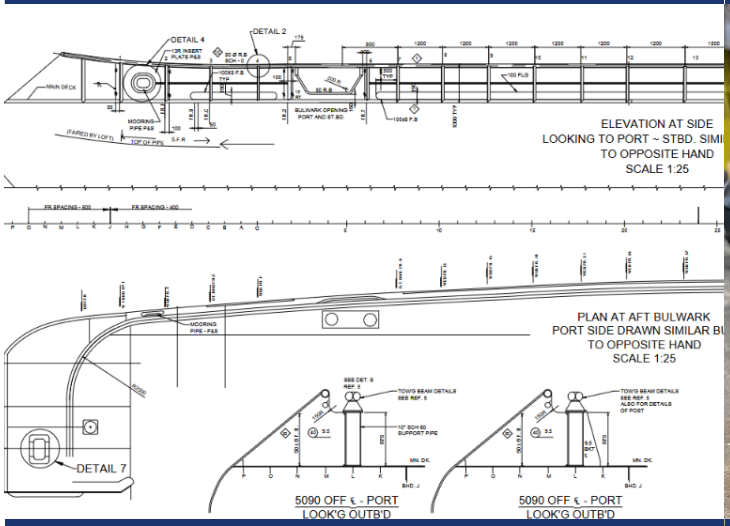
While at Ontario Shipyards I primarily worked on designing and drawing engineering drawings in AutoCAD and SolidWorks for use on the CCGS Terry Fox vessel life extension project. The drawings were used for production and installation of various items on the ships, along with documenting as built conditions. Additionally, I utilized SolidWorks to model part of the ship that was cutout in order to move in the new engines, which was used to visualize the changes to the cutout module and find the center of gravity for lifting. On my last day at my co-op I also got to use the CNC Plasma Cutter and do some welding.



Model of Cutout Section of Ship



Modifying An Assembly Drawing for the Main Engine Coolant Tank




Part of an as Built Engineering Drawing for the Ship Bulwarks

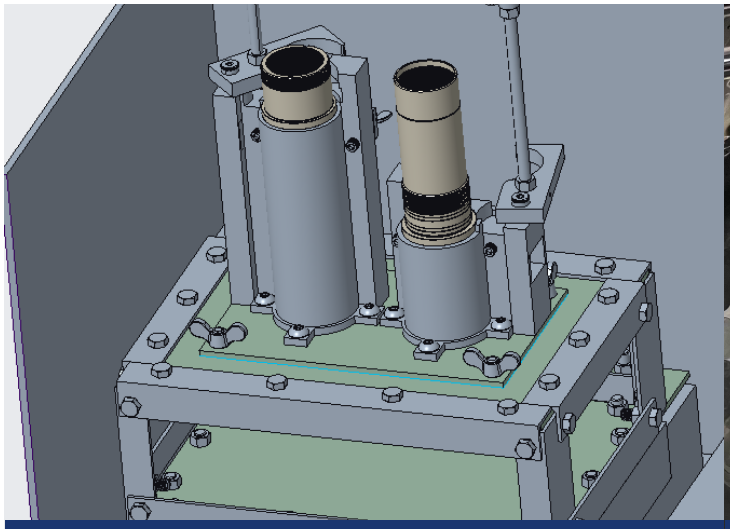
CNC Plasma Cut Maple Leaf that was MIG Welded onto the Base

# Dometic - Co-op Manufacturing Machining

*Creo Parametric, AutoCAD Architecture, Inventor, Navisworks, Python, SQL*

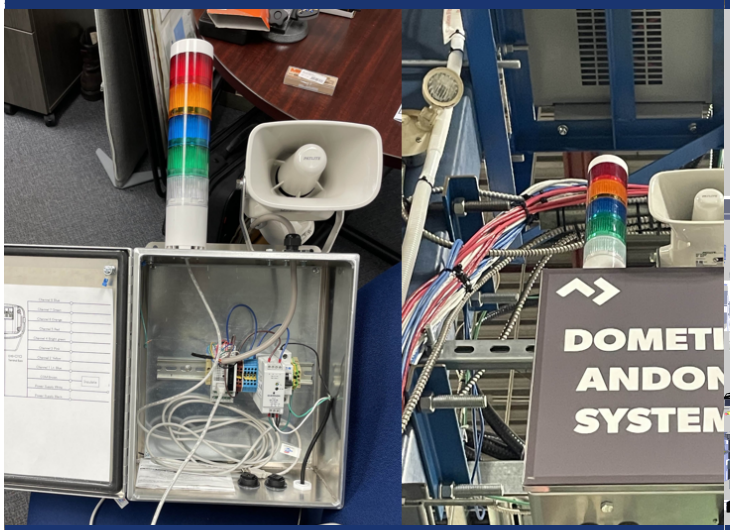
Worked on a wide variety of tasks related to improvements in the factory. Designed jigs in Creo to improve worker productivity, planned out CNC moves, created and implemented announcement boxes, created work standards and promotional posters.

***Rotor Washer Jig*** Created and successfully implemented a custom jig to be used by the workers to wash the internal threads of the rotors after machining and before being nitrated. The jig was designed using water jet cut aluminum and angle, with stainless steel hardware from McMaster Carr.



Rotor Washing Jig Model In Creo

Rotor Washer Implementation




ANDON Boxes

B2 Factory Render

# Takachar - Project Engineer

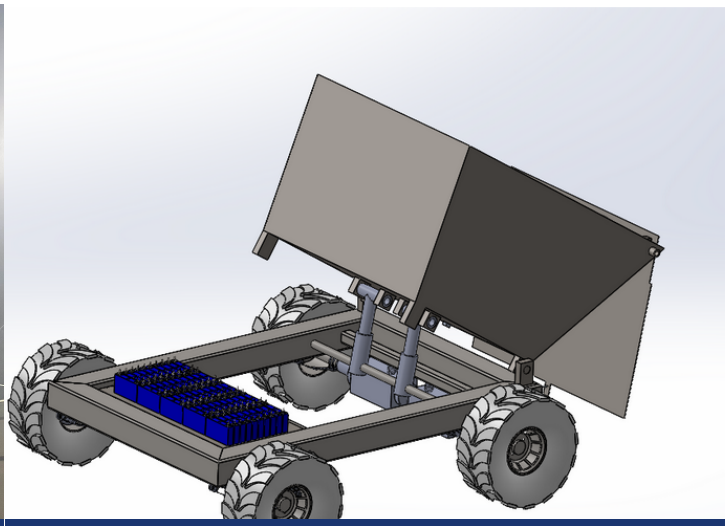
*SOLIDWORKS, MATLAB, Visual Basic For Applications (VBA), Excel*

Experimented with various bioreactor configurations to find the optimal setup by collecting and analyzing reactor data. Created a VBA script that automatically transferred the collected data into Excel to be analyzed. Designed and modeled a remote control electric dump truck to be used to move bio char. Performed regular maintenance on reactor

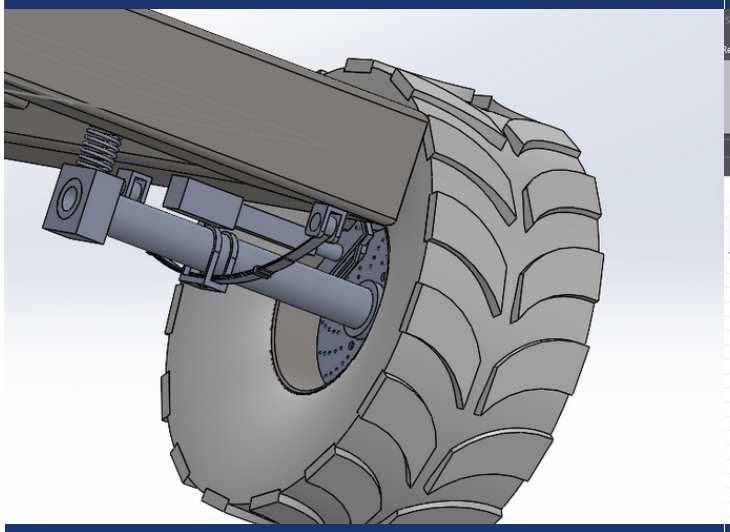
***Remote Control All-Terrain Electric Vehicle:*** Tasked with designing and modeling a remote control all terrain electric vehicle for transportation of biochar between 2 locations. Created detailed SolidWorks models, performed Matlab energy simulations and created a bill of materials

***Automation of Excel using VBA*** VBA Script automatically imports data into Excel from a shared Word Document used to collect data throughout the day. Data is then logged in correct spots in the Excel and results are collected.

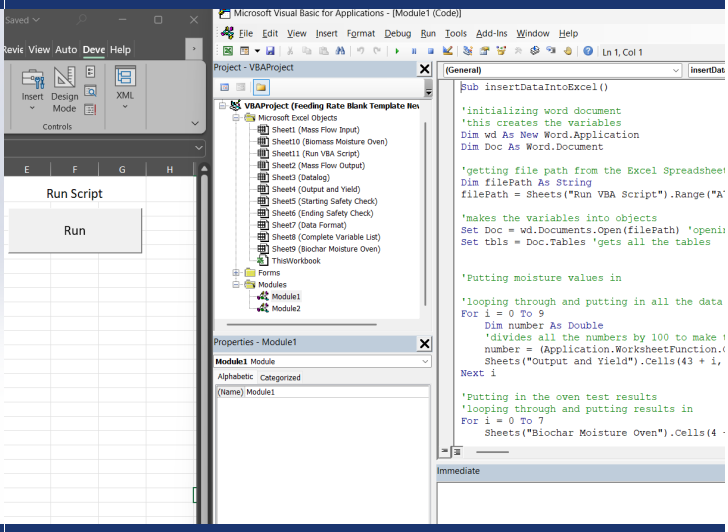
Bioreactor Operating at Night



Overall Model of All Terrain Vehicle



Close Up of Front Wheel of Vehicle



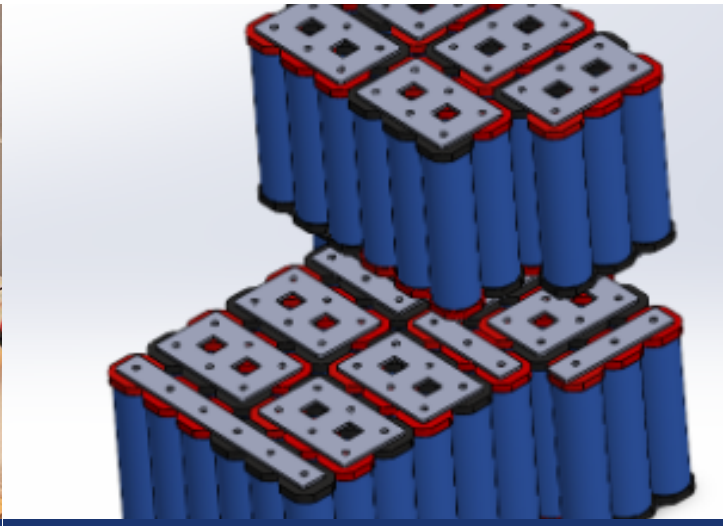

Excel and VBA Script

# ThunderBikes - Battery Team

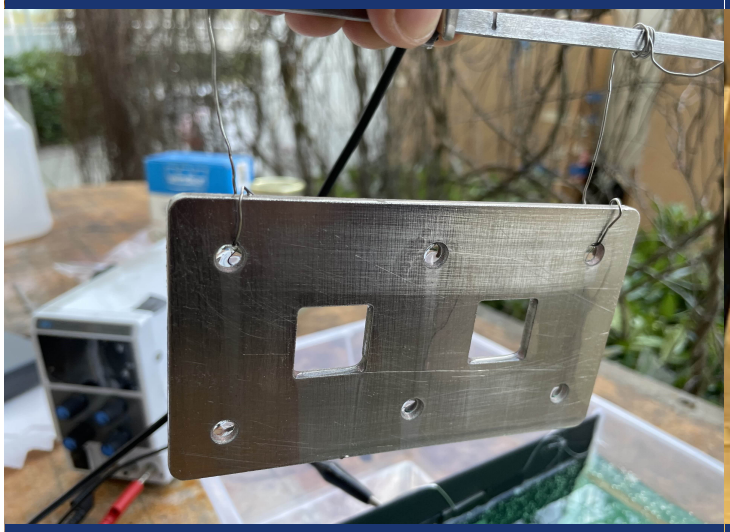
*Battery Design and Manufacturing | SolidWorks, Wiring, Water jet Cutting, Electroplating*

Designed and fabricated a functioning battery pack for an electric motorcycle. Water jet cut copper bus bars to connect battery cells in series and parallel and created 3D printed holding device to hold cells in place. Implemented a battery monitoring system (BMS) to continually monitor cell voltages and protect battery cells. Nickel plated the copper busbars to prevent oxidization.

Assembled Battery Pack



CAD Model of Battery Pack



Nickel Plating Copper Busbars

One of the Battery Modules in its Case

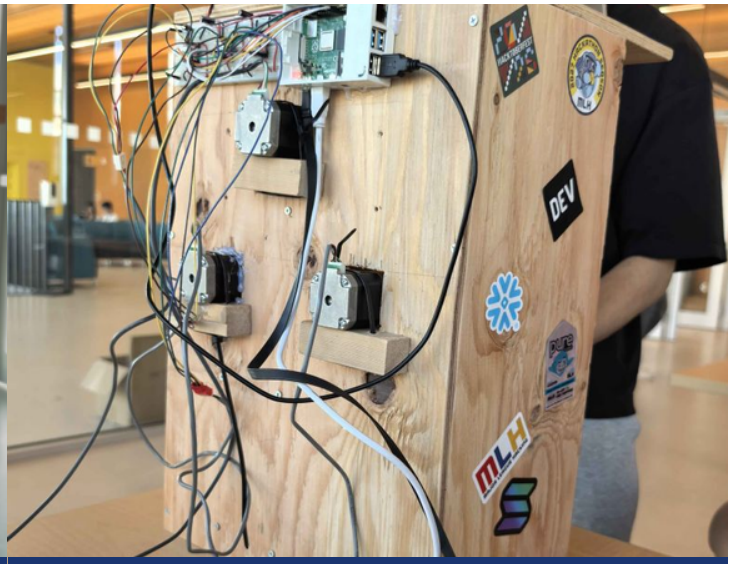
# Automatic Garbage Sorter

*Stormhacks 2026 Hackathon | SolidWorks, 3D Printing, Assembly, Computer Vision, Servo Motors*

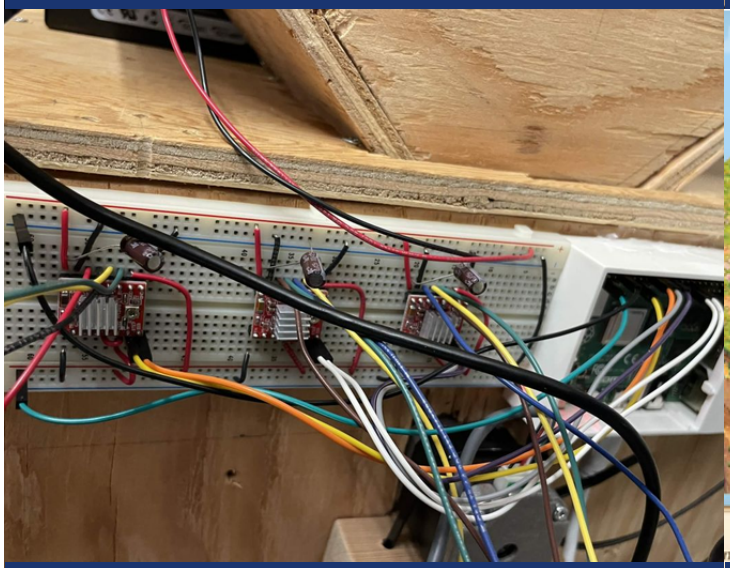
Designed and created an autonomous trash-sorting robot that incorporated a game to encourage waste storage. The device utilized computer vision powered by Gemini to automatically detect and categorize garbage which was then sorted through via a series of platforms powered by servo motors. Based on the garbage placed into the device resource points were awarded which could then be used in the game to build and upgrade a virtual city. I primarily worked on the structural parts and the mechanism for the garbage sorting.

**You can see our submission here:**
https://devpost.com/software/sustanability-promoting-trash-sorting-city-builder-game
https://github.com/Dingxi918/TrashRoyale

View of Our Device From the Front




Hardware for Our Device



Servo Motor Controllers

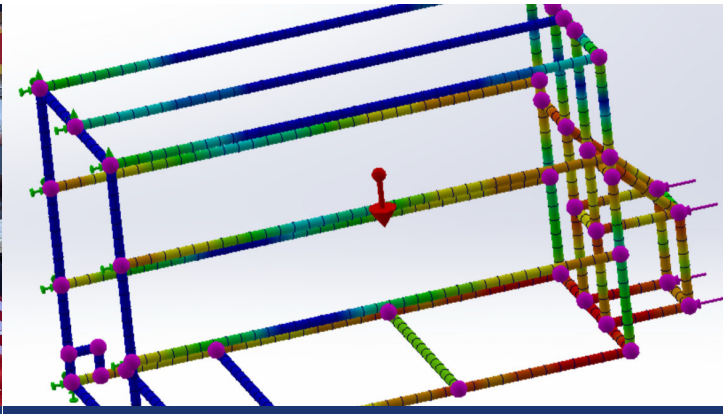
Virtual City Game

## UBC Concrete Toboggan - Super-structure

*Superstructure Sub-team Lead | SolidWorks, FEA, Welding, Jigging, Composites*

Designed and manufactured a roll cage using aluminum capable of withstanding impacts from multiple crash scenarios. Validated roll cage integrity using FEA. Created polycarbonate and fiberglass panels and other safety features to protect riders from external hazards.

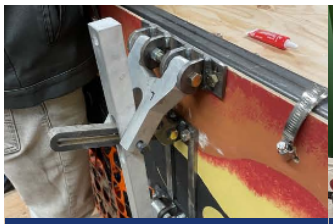
Toboggan on Race Day



Running FEA on Roll Cage

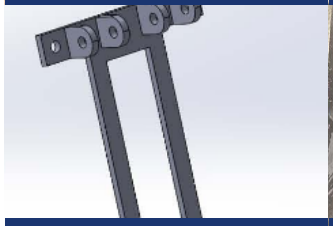
Welding Roll Cage

Making Fiberglass Panels



Final Braking Assembly

Machining Parts



Mounting Plates

Waterjet Cut Pieces

## UBC Concrete Toboggan - Braking

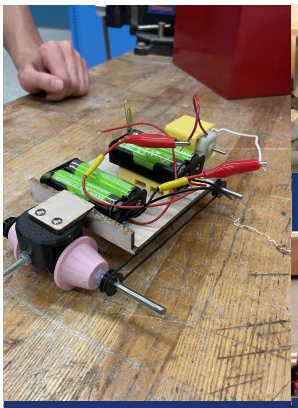
*Braking Sub-team | SolidWorks, Manufacturing*

Designed and ran FEA on mounting plates for the braking system. Water jet cut mounting plates and tabs for the braking apparatus. Manufactured and assembled braking apparatus. Created engineering drawings for parts.

## Locomotive Design Competition

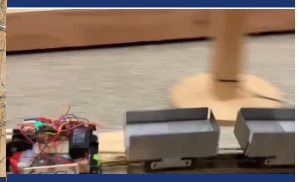
*MECH 223 | SolidWorks, Prototyping, Power Transmission*

Designed and manufactured a locomotive to compete in 2 class wide competitions. Locomotives needed to be fast, able to climb and descend steep grades and pull large payloads. Designed belt tensioner for timing belt power transmission. Competed and won first place in both competitions.



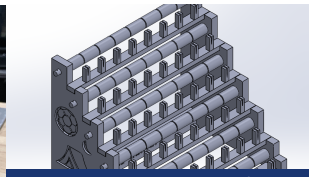
Early prototype of first locomotive

Belt tensioner device on second locomotive

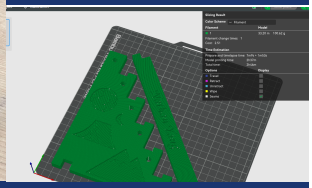


Second locomotive in competition

Finished Assembly



Isometric View of Assembly in CAD



3D Printing Preview

## Thread Spool Holder

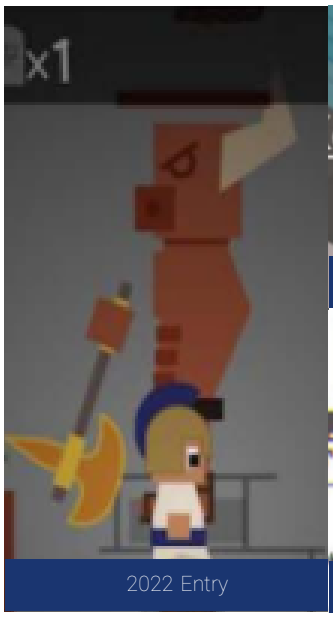
*Personal Project | SolidWorks, 3D Printing*

Created and designed a custom 3D printed assembly for holding a variety of small coloured thread spools. Underwent multiple design iterations that reduced 3D print time by 400%. Incoporated stylistic features to the design.

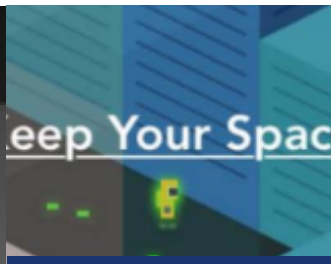
## JS13KGames Competition

*Web Game Dev | JavaScript, HTML5, CSS, Optimization, Algorithms*

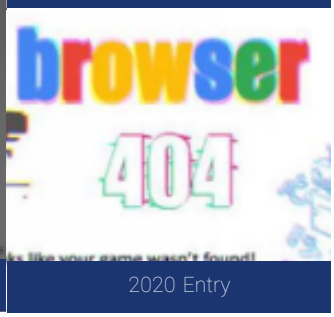
Participated in the annual JS13k challenges for 2020, 2021 and 2022. Created fully functional browser games within a strict 13kB file size limit. The games improved over the years, demonstrating evolution from basic scripting to advanced custom engine architecture. Implemented game logic and path finding algorithms including use of the A* for the 2023 entry.




2022 Entry




2021 Entry




2020 Entry

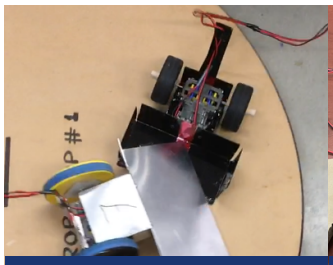
## MECH 2 Levitating Magnet Device

*Machining, Circuits, Soldering, Assembly*

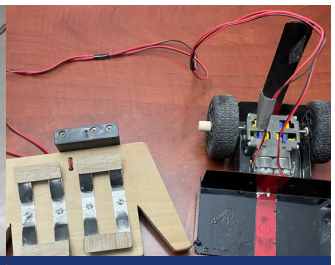
As part of the Mech 2 technical skills practicum, I manufactured and assembled this levitating magnet device. The device works by increasing or decreasing the strength of the magnet field produced by the solenoid at the top, based on feedback from the hall effect sensor in the circuit board. The body of the device is made from bent sheet metal parts and certain components were custom made by using a milling machine or lathe. The circuit board components were soldered on by me, and then assembled into the overall device.

Device Without Magnet

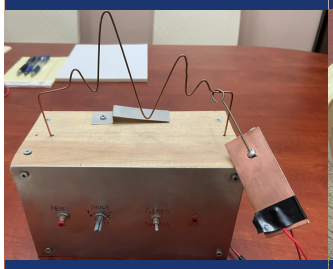
Device Levitating the Magnet



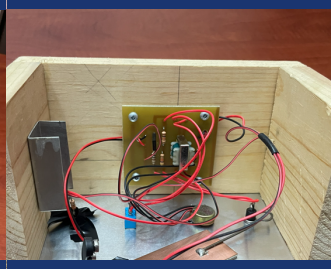
Sumo Robots Battling



Robot and Controller



Shocking Game



Shocking Game Electronics

## High School Engineering and Electronics

*Robotics, Electronics, Soldering*

Designed and fabricated a "Sumo" robot for a class wide competition. Utilized sheet metal tools to create chassis and assembled an electronic controller to control the robot. Won first place in the class. Created various circuits using PCBs and soldered components, including a shocking game and alarm device.